INVESTMENT PORTFOLIO MANAGEMENT: PROJECT 1

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SECTION-1

1(a) Time Series Analysis of Closing Prices

The time series plots show the monthly closing prices of Exxon, Shell, JPMorgan, Wells Fargo, and SPY. The x-axis represents observation numbers from 1 to 71, where each observation matches to one month between January 2018 and December 2023.

Similar price movements are shown by Exxon and Shell stocks throughout the given period. Their prices are stable as indicated by observations 1 to 21 (2018-2019). It then sharply decreases between observations 28 to 35 (2020). Starting with observation 35, the stocks are constantly on the rise, as shown by observations 42 to 71 (2021-2023), signifying the improving state of the energy sector. For both Exxon and Shell, the main turnaround is shown by observations 28 to 35 (2020).

In the JPMorgan data, there is an upward trend from the early stages of the data, i.e., from observations 1 to 21. However, there is an observed decline from 28 to 35. Later on, there is an increased fluctuation in stock prices, unlike before the decline.

The fluctuations in the case of Wells Fargo can be seen as relatively high compared to JPMorgan. The fall in stock values from observations 28 to 35 is even sharper, although a relative increase in stock values can be seen afterward, even though that is weaker and less consistent.

The ETF SPY has a strong uptrend throughout the period, apart from the decline between observations 28 and 35. Even this decline is followed by a stronger and smoother uptrend compared to the stocks themselves after observation 35.

Overall, all series show a common turning point around observations 28–35, showing us the presence of a market-wide shock affecting all securities.

1(b) Time Series Analysis of Monthly Returns

From the time series plots for returns, all securities fluctuate around zero returns, which shows there is no trend in returns. For the beginning period of returns, there is a period of low fluctuations for observations 1-21. Afterwards, around observations 28-35, there is a period of high volatility, as spikes in returns occur.

Exxon and Shell have a wider range of fluctuations than the finance stocks.

The variations for Wells Fargo are more extreme than those for JPMorgan, which implies higher risk.

As shown by the blue line in Figure 1, the SPY returns data series can be said to be smoother compared to that of individual stocks; however, volatility is still evident in data points 28 to 35.

Are outliers present?

Yes, outliers are present. This is observed because most of the observations between 28 and 35 show unusually positive and negative returns.

2. Descriptive Statistics of Returns (Table 1)

Table 1: Descriptive Statistics and Performance Comparison

Variable	Arithmetic Mean	Geometric Mean	Standard Deviation
XOM_RET	0.642528	0.5716	9.60304
WFC_RET	0.5339937	0.24323	9.56568
JMP_RET	0.841021	0.57865	7.73874
SHEL_RET	0.210722	0.23502	7.91098
SPY_RET	0.875561	0.33951	5.25460

Table 1 presents the arithmetic mean, geometric mean, and standard deviation of the monthly returns of Exxon (XOM), Shell (SHEL), JPMorgan (JMP), Wells Fargo (WFC), and SPY from January 2018 to December 2023.

Return comparison:

SPY has the highest arithmetic mean at 0.8756, which means it is the strongest in terms of average performance. JPM also performs well as it has a relatively high arithmetic mean of 0.8410. Shell has the lowest arithmetic mean, 0.2107.

Risk Comparison:

The investment with the lowest standard deviation is SPY, at 5.2546, which means that SPY is the least volatile stock. XOM and Wells Fargo are more volatile since their standard deviation is more than 9.5.

Overall Conclusion: SPY performs better than other stocks in terms of returns and risk. Therefore, it can be concluded that SPY is a better investment option than other stocks.

3. Correlation Between Securities

Correlation Between Securities

	XOM_RET	**WFC_RET**	**JMP_RET**	**SHEL_RET**
XOM_RET	-	0.651	0.570	0.770
WFC_RET	0.651	-	0.789	0.655
JMP_RET	0.570	0.789	-	0.601
SHEL_RET	0.770	0.655	0.601	-

This table presents the correlation of the monthly returns of Exxon (XOM), Shell (SHEL), JPMorgan (JMP) and Wells Fargo (WFC) from January 2018 to December 2023.

Correlation of stock prices within the industry:

The Exxon stock prices present the highest correlation with Shell stock prices, with a 0.770 correlation and vice versa. Similarly, the Wells Fargo stock prices exhibit the highest correlation with JPMorgan Chase stock prices, with a 0.789 correlation and vice versa. We can infer that the correlation among the stock prices within an industry is higher than the correlation between industries.

This is the case, as companies share external tailwinds and headwinds within an industry. They are affected by similar industry changes, be it technological advancements or regulatory measures. Investors' perception also causes company stock prices of the same industry to move together.

4. Price Weighted Index Returns (Table 2)

Table 2: Price Weighted Index Returns

Variable	**Arithmetic Mean**	**Geometric Mean**	**Standard Deviation**
PR_WT_RET	0.443758	0.33042	0.27894

Table 2 presents the arithmetic mean, geometric mean, and standard deviation of returns for the price-weighted index created from Exxon, Shell, JPM, and Wells Fargo.

The arithmetic mean return of 0.4438 suggests that, on average, over the sample period, the price-weighted index achieved a moderate average monthly return. The geometric mean with a lower value, of 0.3304, reflects the impact from compounding and volatility on long-term growth.

The low standard deviation of 0.2789 shows that the price-weighted index is highly stable and less volatile as compared to individual stocks. This therefore means that the index benefits from diversification; the overall risk is minimized.

Overall Conclusion

It gives an average return with low risk, making the price-weighted index a more stable investment as opposed to overdependence on individual stocks.

5. Value Weighted Index Returns (Table 3)

Table 3: Value Weighted Index Returns

Variable	**Arithmetic Mean**	**Geometric Mean**	**Standard Deviation**
PVR_WT_RET	2.93597	1.7506	25.2325

Table 3 shows the arithmetic mean, geometric mean, and standard deviation of returns for the value-weighted index containing Exxon, Shell, JPM, and Wells Fargo.

The arithmetic mean return of (2.9360) indicates how well the value-weighted index performed in terms of average return as opposed to individual stocks and the price-weighted index. Although it is declining, the geometric mean return of (1.7506) is still impressive.

However, the standard deviation is very high, which is 25.2325. This implies that the value-weighted index is very volatile and risky. This simply implies that the returns are influenced more by larger companies, resulting in more gains or losses.

Overall Conclusion

If the returns are much more significant, then the level of risk is far too high. This index is generally for investors who want to take risks to gain better returns.

6. Equal Weighted Index Returns (Table 4)

Table 4: Value Weighted Index Returns

Variable	**Arithmetic Mean**	**Geometric Mean**	**Standard Deviation**
PRICE EQUAL_WT_RET	0.437066	0.14424	7.57074

Table 4 shows the arithmetic mean, geometric mean, and standard deviation of returns for the equal-weighted index containing Exxon, Shell, JPM, and Wells Fargo.

The arithmetic mean return of 0.4371 indicates the average performance of the equal-weighted index, where each stock contributes equally to the overall return. This suggests that, on average, the index generated positive returns due to the combined performance of all four stocks.

The geometric mean return of 0.1442, however, is relatively low compared to the arithmetic mean. This indicates that when compounding over time is considered, the overall growth of the index is lower. The gap between the arithmetic and geometric mean suggests variability in returns across periods.

The standard deviation of 7.5707 indicates a moderate level of volatility. This implies that while the index experiences fluctuations, the risk is significantly lower compared to a value-weighted index. Since all stocks are equally weighted, no single large company disproportionately affects the index's movements.

Overall Conclusion:

The equal-weighted index provides more balanced exposure across all companies, resulting in moderate returns with comparatively lower risk. This index is more suitable for investors who prefer diversification and stability rather than extremely high returns accompanied by high volatility.

7. Price Weighted Index vs Value Weighted Index vs Equal Weighted Index

Arithmetic Mean: The value-weighted index has the highest arithmetic mean (2.93597), largely due to its structure, as larger firms receive higher weights and dominate returns. The equal-weighted index and the price-weighted index have the lower arithmetic mean (0.437) and (0.44376) since it reflects the average performance of all firms equally and depend only on stock prices and ignore firm size and market significance ,respectively.

Geometric Mean: The value-weighted index also records the highest geometric mean (1.7506), indicating stronger long-term compounded returns driven bythe consistent performance of large firms. The price-weighted index shows moderate compounded growth (0.33042). The equal-weighted index has the lowest geometric mean (0.14424), suggesting weaker compounding due to uneven returns across firms.

Standard Deviation: The value-weighted index has the highest standard deviation (25.2325), making it the most volatile due to heavy exposure to large firms. The equal-weighted index shows moderate risk (7.57074). The price-weighted index has the lowest volatility (0.27894), though it is less representative of overall market movements.

Which Index Represents the Market Better?

The value-weighted index most closely represents the true market performance. Because larger companies contribute more to the index, which aligns with how real markets behave. It captures the performance of the largest firms which drive market movements. Benchmarks like the S&P 500, FTSE, and NIKKEI are also value-weighted for the same reason.

Overall Conclusion:

While the value-weighted index is more volatile, it better reflects actual market behavior compared to the price-weighted and equal-weighted indices. It's most suitable for investors trying to understand or mirror overall market performance.

Note: Description of Observation Index Used in Time Series Plots

The dataset contains monthly observations from January 2018 to December 2023, a total of 72 observations. In the following Minitab output time series plots, the x-axis is scaled in observation number, not calendar date.

The observation number refers to one month in chronological order. As such, observation 1 represents January 2018 and observation 72 represents December 2023.

To avoid overcrowding, on the x-axis, Minitab displays selected observation numbers (such as 1, 7, 14, 21, …71) as reference points. All analysis and interpretations are therefore based on these observation numbers and their corresponding positions in the sample period.

SECTION-2

1. Exxon & Shell

The minimum variance portfolio allocates higher weight to Shell (89.23%) and lower weight to Exxon (10.77%) because Shell has lower volatility. The portfolio standard deviation (0.0788) is lower than both Exxon (0.0960) and Shell (0.0791), showing diversification benefits. The portfolio return (0.00257) lies between individual returns as expected. While Exxon has the highest individual Sharpe ratio, the portfolio improves the overall risk-return trade-off compared to holding Shell alone.

TABLE 5: Exxon & Shell

Stock	Weights	Avg Return	Std Dev	Sharpe Ratio
Exxon	0.107730	0.0064253	0.0960304	0.0514504
Shell	0.892270	0.0021072	0.0791098	0.0078713
MVP	1	0.0025724	0.0788323	0.0138002

2. Exxon & JPM

The Exxon-JPM portfolio provides stronger diversification than the Exxon-Shell portfolio. In the Exxon-Shell case, the portfolio risk only drops slightly from 0.0791 (Shell) and 0.0960 (Exxon) to 0.0788 because both are oil sector stocks and move similarly. However, in the Exxon-JPM portfolio, the risk decreases more meaningfully to about 0.0773, lower than both Exxon (0.0960) and JPM (0.0774). The Sharpe ratio also improves from 0.0138 in the Exxon-Shell portfolio to about 0.0829 in the Exxon-JPM portfolio. This shows that combining stocks from different industries (oil and banking) gives better diversification and a better overall risk-return trade-off than combining similar stocks.

TABLE 6: Exxon & JPM

Stock	Weights	Avg Return	Std Dev	Sharpe Ratio
Exxon	0.260114	0.0064253	0.0960304	0.0514504
JPM	0.739886	0.0084102	0.0773874	0.0894939
MVP	1	0.0078939	0.0743833	0.0861672

The portfolio weights change significantly after Covid. In the pre-Covid period, the portfolio mainly invests in Shell (72.92%) and less in Exxon (27.08%), reflecting Shell's lower risk. In the post-Covid period, Exxon receives a negative weight (-30.08%) while Shell exceeds 100% (130.08%), meaning the model suggests short-selling Exxon to minimize risk. This happens because the two oil stocks moved much more similarly after Covid, reducing diversification benefits. As correlation increased, the portfolio required a short position to achieve minimum variance.

SECTION-3

3.1 Exxon & Shell (Pre-Covid & Post-Covid)

The portfolio weights change significantly after Covid. In the pre-Covid period, the portfolio mainly invests in Shell (72.92%) and less in Exxon (27.08%), reflecting Shell's lower risk. In the post-Covid period, Exxon receives a negative weight (-30.08%) while Shell exceeds 100% (130.08%), meaning the model suggests short-selling Exxon to minimize risk. This happens because the two oil stocks moved much more similarly after Covid, reducing diversification benefits. As correlation increased, the portfolio required a short position to achieve minimum variance.

TABLE 7.1: Exxon & Shell (Pre-Covid)

Stock	Weights	Avg Return	Std Dev	Sharpe Ratio
Exxon	0.270828	-0.0030734	0.0838576	-0.0421897
Shell	0.729172	-0.0068258	0.0944827	-0.0922826
MVP	1	-0.0058095	0.0820296	-0.0819497

TABLE 7.2: Exxon & Shell (Post-Covid)

Stock	Weights	Avg Return	Std Dev	Sharpe Ratio
Exxon	-0.300794	0.0250269	0.0983212	0.228056
Shell	1.30079	0.0196010	0.0670866	0.253358
MVP	1	0.0179690	0.0648749	0.236837

3.2 Exxon, Shell, JPM & Wells Fargo

The alpha represents the average abnormal return when the market excess return is zero. Exxon (α = 0.0073) and JPM (α = 0.00736) have positive alphas, suggesting they slightly outperform the market on average, while Shell (α = 0.00134) shows a very small positive abnormal return and Wells Fargo (α = −0.0010) shows a small negative abnormal return. However, all alpha p-values are high (> 0.10), so none of these abnormal returns are statistically different from zero — meaning there is no reliable evidence that any stock consistently outperforms or underperforms the market.

The beta measures sensitivity to market movements. All stocks have negative betas: Exxon (β = −0.234), Shell (β = −0.081), JPM (β = −0.195), and Wells Fargo (β = −0.169). This suggests that during this sample period, these stocks tended to move slightly opposite to the market. However, the beta p-values are also large (> 0.10), so the relationship is not statistically significant — we cannot conclude the market actually drives these movements.

The R^2 values are extremely low: Exxon 1.62%, Shell 0.29%, JPM 1.77%, and Wells Fargo 0.86%. This means less than 2% of the variation in stock excess returns is explained by market excess returns. Therefore, most of the variation is due to firm-specific factors rather than systematic market risk.

Overall, since both alpha and beta are insignificant and R^2 is near zero, none of the stocks are closely aligned with the market, and their returns are primarily driven by idiosyncratic influences rather than market performance.

TABLE 8: Exxon, Shell, JPM & Wells Fargo

Stock	Alpha	Beta	R^2
Exxon	0.0073	-0.234	0.0162
Shell	0.00134	-0.081	0.0029
JPM	0.00736	-0.195	0.0177
Wells Fargo	-0.0010	-0.169	0.0086

3.3 Wells Fargo & Shell (Least Market Dependent Stocks)

The minimum variance portfolio assigns a higher weight to Shell (0.7566) and a lower weight to Wells Fargo (0.2434), indicating Shell contributes more stability to the portfolio. Wells Fargo has a slightly negative average return (-0.00094) and the highest volatility (0.09553), which also gives it a negative Sharpe ratio (-0.0249). Shell, on the other hand, has a positive average return (0.00214) with lower risk (0.07968) and a positive Sharpe ratio (0.00870), making it the stronger asset in the pair.

After combining them, the portfolio average return becomes 0.00139 and the standard deviation reduces to 0.07764- lower than both individual stock - showing diversification benefits. However, the portfolio Sharpe ratio (-0.00071) is approximately zero, meaning the improvement in risk reduction does not translate into meaningful risk-adjusted performance.

Overall, even though these two stocks were chosen because they are least dependent on the market, the portfolio mainly reduces risk rather than improving returns, demonstrating that low market correlation alone does not guarantee a good Sharpe ratio; return quality also matters.

TABLE 9: Wells Fargo & Shell (Least Market Dependent Stocks)

Stock	Weights	Avg Return	Std Dev	Sharpe Ratio
WFC	0.243392	-0.0009357	0.0955300	-0.0249136
Shell	0.756608	0.0021373	0.0796806	0.0086972
MVP	1	0.0013894	0.0776432	-0.0007071

CITATIONS

- Class Notes
- Running Notes
- Generative AI